ML MINI PROJECT · AM321 & AM312

Can a computer guess your doodle?

SVMs vs CNNs on hand-drawn sketches from Google's Quick, Draw! dataset

28 × 28

pixels is all it gets

THE PROBLEM

Sketch in, label out

A messy doodle

28 × 28 grayscale pixels, drawn in 20 seconds by a stranger

A model

Classic ML (SVMs) or a CNN, trained on labelled examples

A guess

"That's a banana." Right or wrong.

Our question: which model reads doodles best, and when does it matter?

Three difficulty levels

3

classes

banana, apple, cat

10

classes

the 3 above, plus dog, car, bicycle, airplane, chair, tree, fish

50

classes

everyday objects, airplane to flower

28 × 28 grayscale

20,000 per class

80 / 10 / 10 split

6,000 train images each

THE CONTENDERS

Two teams, one fair fight

Team Classic ML

Fast, simple, no deep learning

- Logistic Regression
- Linear SVM
- RBF SVM
- Polynomial SVM
- Sigmoid SVM

Team CNN

Learns its own features straight from pixels

- CNN V1 (the full model)
- CNN V2
- CNN V3
- CNN V4 (the smallest)
- Each one on grayscale and on black-and-white pixels

Fair fight rule: same 6,000 training images for everyone

RESULTS

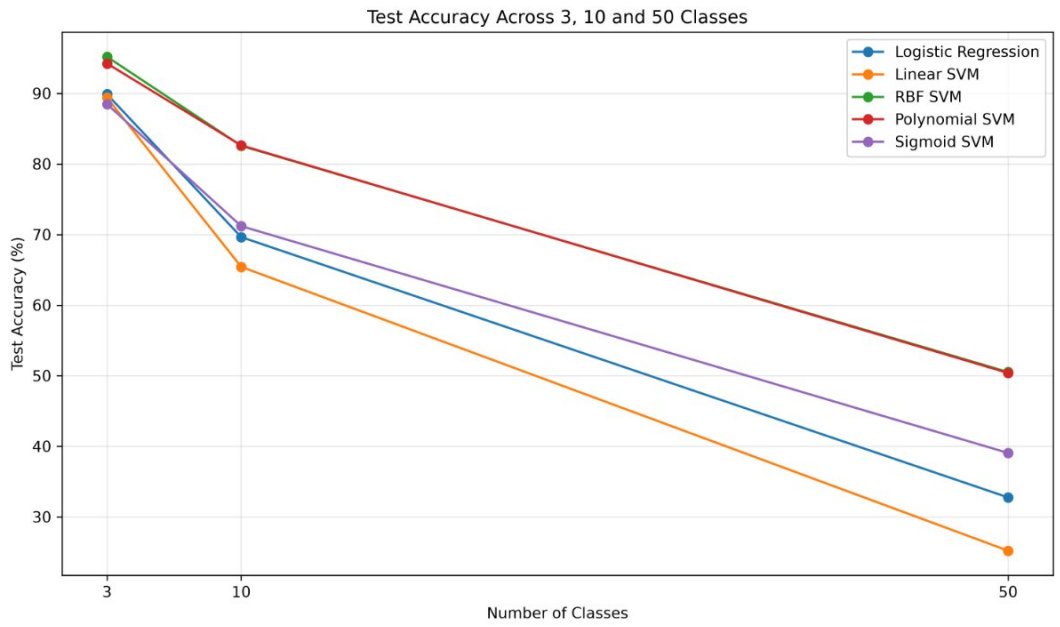
Results: 10 classes

Test accuracy. The CNN is 3.7 points ahead of the best SVM.



RESULTS

50 classes: everything drops



50.5%

best score at 50 classes (RBF SVM)

Random guessing gets 2%

Polynomial SVM is right behind at 50.4%. Linear SVM crashes to 25.2%.

THE TRADE-OFF

CNN wins accuracy, SVM wins speed

+3.7 pts

more accurate

CNN V1 86.34% vs best SVM 82.63% at 10 classes

~6×

slower to train

58.1 s vs 9.8 s at 10 classes (31.2 s vs 5.5 s at 3 classes)

Need the best accuracy? CNN. Need fast and good enough? RBF SVM.

ERROR ANALYSIS · 50-CLASS RBF SVM

Where it breaks



Hardest: dog, dragon, bear

F1 scores of only 0.14, 0.18 and 0.20

Easiest: apple, envelope, candle

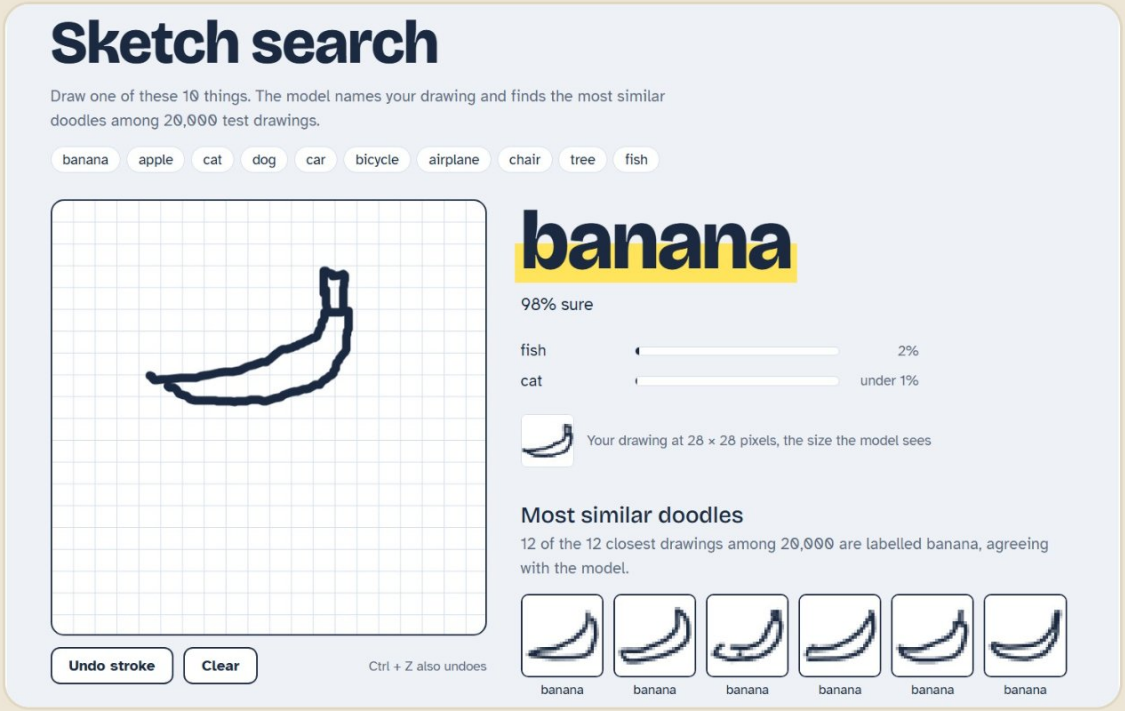
F1 scores of 0.83, 0.81 and 0.77

Classic mix-ups

feather → carrot (318), arm → banana (283), dolphin → bridge (282), calendar ↔ calculator (266 and 243)

LIVE DEMO

Draw it, we guess it



1 · Draw

Sketch one of the 10 things in the browser.

2 · Guess

The CNN names it and shows how sure it is.

3 · Find twins

It pulls the 12 most similar test doodles. They vote too.

TAKEAWAYS

The short version

1. CNNs win on accuracy: 86.3% at 10 classes vs 82.6% for the best SVM.

2. RBF SVM is the best classic model, and about 6× faster to train than the CNN.

3. More classes means a harder game: the best model we tested reaches only 50.5% at 50 classes.

Next: CNNs on 50 classes, trained on all 16,000 images per class

Questions?